

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	26 June 2026
Team ID	Solo
Project Name	ShopEZ - E-Commerce Application
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration & Authentication	Registration/login through form (name, email, password); Password hashed with bcrypt before storing; JWT issued on successful login
FR-2	Product Catalog Browsing	Live search by product name/keyword; Category filters (e.g., "Home & Kitchen") with URL-parameter sync; View product details with specifications and reviews tabs
FR-3	Shopping Cart Management	Add/remove/update item quantity in cart; Live price calculation on every change; Cart auto-syncs to the database for the logged-in user
FR-4	Checkout & Order Placement	3-step checkout: shipping address → billing details → payment; Payment options: Card, UPI, Cash on Delivery; Order confirmation and order ID generation
FR-5	Customer Profile Management	Edit and save shipping/billing addresses; View order status history (Pending/Processing/Shipped/Delivered)
FR-6	Seller Inventory Management	Add new product via form modal; Edit existing product details and price; Delete a product from the catalog

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Clean, boutique-style UI with intuitive navigation, toast notifications (React Hot Toast) for instant feedback on actions like add-to-cart or order updates
NFR-2	Security	JWT-based stateless authentication, bcrypt password hashing, environment secrets kept out of Git via .gitignore, and no exposed credentials in the repository
NFR-3	Reliability	Consistent cart-to-database syncing and accurate order-status transitions, with a corrected database seeder ensuring valid password hashes for all users
NFR-4	Performance	Fast page loads via Vite-optimized React build; efficient MongoDB queries through Mongoose for catalog search and filtering

NFR-5	Availability	Hosted on Render with a unified render.yaml blueprint keeping client and server deployed together, minimizing downtime during updates
NFR-6	Scalability	Cloud-hosted MongoDB Atlas database and a modular Express REST API that can handle growing product catalogs, users, and order volume without architecture changes